

Case Study Analysis

Autonomous Agents in Industry 4.0
Course: ITAI 3377 | Assignment: A09

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Part 1: Case Study Analysis

Introduction

Autonomous agents are computer systems or physical machines that can sense what is around them and make their own decisions to get a job done, and they have become increasingly important in Industry 4.0 as factories rely more on connected technology and real-time data. Industry 4.0 refers to the fourth industrial revolution, which brings together Cyber-Physical Systems, artificial intelligence, and automation so that machines can communicate with each other and work with less human involvement. Traditional automated guided vehicles have had a hard time fitting into this new type of environment because they follow set paths using physical markers on the floor and depend on a central system to tell them where to go, which makes them slow to respond when things change. Leo Hjulstrom's 2022 thesis from KTH Royal Institute of Technology looks at how reinforcement learning can be used to make these vehicles smarter, so they can figure out how to move around and manage their own battery life through trial and error rather than being told exactly what to do at every step.

Implementation

The system Hjulstrom built is made up of three agents working together inside a simulated 10x10 grid warehouse, where each agent has to carry objects from one spot to another while also keeping an eye on its battery and heading to a charging station before it runs out of power. The project was written in Python using the TensorFlow and TF-Agents libraries, and the Double Deep Q-Network algorithm was chosen because it had already been shown to work well for this type of navigation task and trains faster than other options like DDPG. Each agent received four pieces of information at every step, which were its own position in the grid, the position of its next target square, its current battery level, and the state of the four squares directly around it, and the agent would then choose to move up, down, left, or right. The reward system gave agents a score of positive 100 for reaching a target, negative 70 for crashing into a wall or another agent, negative 100 for running out of battery, and a small negative 1 for every step taken to push them toward finishing tasks without wandering around. The system trained for 200,000 episodes total, and the number of tasks each agent had to

complete per episode went up by one every 30,000 episodes, which helped the agents learn battery management at the same time as they were learning to navigate.

Benefits

The final test showed strong results, with all three agents finishing all 300 tasks across the whole run without any crashes or battery failures, and they averaged only 2.59 extra steps per task on top of the shortest possible route, which accounts for the times they needed to stop and recharge or move around another agent. Getting rid of physical floor markers is one of the biggest practical wins here, since those markers are expensive to maintain and take the vehicles offline whenever they need to be cleaned or replaced, problems that this learning-based approach avoids entirely. Because each agent makes its own decisions without a central controller telling it what to do, the system as a whole keeps working even if one agent breaks down, which makes it more reliable in real use. The agents also managed to learn on their own how to balance getting tasks done fast while still leaving enough time to recharge, which is a tricky balance that would take a lot of careful hand-written rules to recreate using older methods.

Challenges

The study does have some clear limits that are important to be honest about. The whole system was only tested in a simple 2D simulation with one fixed layout, so it is not clear how well it would hold up in a real factory or in a warehouse with a different setup. The agents had no way to talk to each other, meaning there was no system in place for deciding which agent should take which task or for checking whether a charging station was already being used, which could cause problems in a busier environment with more agents. The battery in the simulation also behaved in an unrealistic way, losing charge at the exact same rate every step and refilling instantly when the agent arrived at a station, while real batteries slow down over time and take a while to charge back up. The paper also briefly mentions that around 47 percent of jobs in the United States could be at risk from automation, which is worth thinking about more seriously since systems like this one are designed to replace the kind of work that people currently do in warehouses and factories.

Future Implications

One of the most useful next steps for this research would be giving agents the ability to communicate with the charging stations, so the stations could let agents know whether they are free, how long the wait is, and which one would be the best to go to, since that kind of information sharing could make the whole system run more smoothly when more vehicles are added. Testing the agents in different warehouse layouts would also help show whether what

they learned in one setup carries over to others, and since related research has already shown that agents trained in a simulation of a real factory can be moved to actual hardware without starting over, this approach could be used in buildings that are still being built or in factories going through changes. As more devices and products in Industry 4.0 environments start sharing data in real time, the agents will have more information to work with, which could allow them to do things like reroute around unexpected obstacles, flag when they need maintenance, or better coordinate with people working in the same space. Hjulstrom's work shows that this kind of self-learning vehicle system is within reach, as long as future versions close the gap between what works in a simulation and what is needed in a real working environment.

Part 2: Personal Reflection

Reading this case study changed how I think about reinforcement learning, because I used to see it as something that works well in demos but is not quite ready for real problems, and seeing the agents finish 300 tasks with zero errors made that view harder to defend. The part that stuck with me the most was how important the reward system is, since the way points were given and taken away for each action is really what taught the agents to behave in a useful way, which means that designing one of these systems is just as much about clearly deciding what you want the agent to do as it is about knowing the technical side of machine learning. That feels like a lesson that applies beyond just robotics, because in any situation where a system is learning from feedback, getting the feedback right matters more than almost anything else.

The part about job automation also stayed with me, mostly because the paper only spends a couple of sentences on it before moving on, and that felt like it did not match the size of the issue since systems like this one are built to take over the kind of work that real people are doing right now in warehouses and factories all over the world. The research itself is solid and the benefits are real, but as these tools get closer to being used at scale, the conversation about what happens to the workers being replaced needs to be taken just as seriously as the conversation about efficiency and performance numbers.